

8. Decision Tree Prediction Using WEKA

Your PDF specifically asks for categorical-data prediction through WEKA.

Steps

1. Open **WEKA** → **Explorer**.
2. Go to **Preprocess**.
3. Load any suitable categorical dataset.
4. Click **Classify**.
5. Click **Choose**.
6. Select:

Trees → J48

7. Select **10-fold cross-validation**.
8. Click **Start**.
9. Record the accuracy.
10. View the generated tree.
11. Take a screenshot.

Program Code shopping.arff

```
@relation shopping

@attribute Milk {yes,no}
@attribute Bread {yes,no}
@attribute Butter {yes,no}
@attribute Eggs {yes,no}

@data
yes,yes,no,yes
yes,yes,yes,no
yes,no,yes,yes
no,yes,yes,yes
yes,yes,yes,yes
no,yes,no,yes
```

OUTPUT :

The screenshot shows the Weka Explorer interface with the 'Associate' tab selected. The 'Apriori' algorithm is chosen, and the 'Start' button has been clicked. The 'Result list' on the left shows the execution time as 09:56:21. The main window displays the 'Associator output' for the Apriori algorithm.

Associator output

```
Instances: 6
Attributes: 4
Milk
Bread
Butter
Eggs

==== Associator model (full training set) ====

Apriori
Minimum support: 0.55 (3 instances)
Minimum metric <confidence>: 0.6
Number of cycles performed: 9

Generated sets of large itemsets:

Size of set of large itemsets L(1): 4
Size of set of large itemsets L(2): 6

Best rules found:
1. Eggs=yes 5 ==> Bread=yes 4 <conf: (0.8) > lift: (0.96) lev: (-0.03) [0] conv: (0.42)
2. Bread=yes 5 ==> Eggs=yes 4 <conf: (0.8) > lift: (0.96) lev: (-0.03) [0] conv: (0.42)
3. Milk=yes 4 ==> Bread=yes 3 <conf: (0.75) > lift: (0.9) lev: (-0.06) [0] conv: (0.33)
4. Butter=yes 4 ==> Milk=yes 3 <conf: (0.75) > lift: (1.13) lev: (0.06) [0] conv: (0.67)
5. Milk=yes 4 ==> Butter=yes 3 <conf: (0.75) > lift: (1.13) lev: (0.06) [0] conv: (0.67)
6. Milk=yes 4 ==> Eggs=yes 3 <conf: (0.75) > lift: (0.9) lev: (-0.06) [0] conv: (0.33)
7. Butter=yes 4 ==> Bread=yes 3 <conf: (0.75) > lift: (0.9) lev: (-0.06) [0] conv: (0.33)
8. Butter=yes 4 ==> Eggs=yes 3 <conf: (0.75) > lift: (0.9) lev: (-0.06) [0] conv: (0.33)
9. Bread=yes 5 ==> Milk=yes 3 <conf: (0.6) > lift: (0.9) lev: (-0.06) [0] conv: (0.56)
10. Eggs=yes 5 ==> Milk=yes 3 <conf: (0.6) > lift: (0.9) lev: (-0.06) [0] conv: (0.56)
```